An unconditional bound for a Möbius-weighted correlation sum in fixed residue classes, with two consequences for the Huang–Li reduction of Goldbach to Elliott–Halberstam

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Abstract

Huang and Li [2] show that the binary Goldbach conjecture for large even N follows from EH together with a Möbius-twisted variant EH_μ whose levels sum to more than 1; by Bombieri–Vinogradov their Corollary 1 reduces this to $EH_\mu(N^{\theta'})$ alone for a single $\theta' > 1/2$. Their proof consumes EH_μ at exactly two places, the functionals they call $E_3(\alpha)$ and $E_4(\alpha)$. We prove unconditionally (Theorem 1) that the Möbius-weighted correlation sum underlying E_4 is $\ll_A N(\log N)^{-A}$ for every $A > 0$, uniformly in the truncation point; consequently (Corollary 2) the E_4 consumption of EH_μ is unnecessary and the entire EH_μ demand of the Huang–Li argument collapses to the single scalar $E_3(\alpha)$. We then show (Theorem 3) that the corresponding statement for E_3 — the same sum with the weight $\log k$ — is not merely hard but, by an unconditional identity, *equivalent* to Huang–Li's own equation (22): it therefore already yields binary Goldbach for large even N, and yields the asymptotic $\tilde{r}(N) \sim \mathfrak{S}(N)N$ exactly when $\sum_{n<N} \Lambda(n)\mu(N-n) = o(N)$. The root cause is $\mu * \log = \Lambda$. We then locate the true strength of the surviving demand: Goldbach needs only a one-sided bound on E_3 (Proposition 5), at a threshold of order N for almost all N, and hence only a constant-factor bound on exactly the object EH_μ bounds (Proposition 6). Finally we record a defect in [2], first observed by S. Zheleznov: equation (18) drops an n-dependent constraint present in the definition of $S_2(\alpha)$. The missing term Δ is exhibited and shown harmless under the hypotheses already assumed (Section 7). The authors have since corrected the manuscript by keeping the constraint instead, so that Δ does not arise there; the two formulations differ by exactly Δ's shape, and Theorem 1 together with that estimate gives the bound against the corrected one (Proposition 20).

We state plainly what is not claimed. Theorem 1 removes only the part of the demand that carries no Goldbach content, and the net progress toward the Goldbach conjecture is zero. We also state plainly what is standard: the ingredients of Theorem 1 are classical and its mechanism — a divisor switch that moves the work onto a short variable, followed by Bombieri–Vinogradov — is ordinary practice. What is offered is the application, together with the corrections it turns out to require.

1 Introduction

1.1 The Huang–Li reduction

Let N be a large even integer, Λ the von Mangoldt function, μ the Möbius function, and

$$\tilde{r}(N) = \sum_{n<N} \Lambda(n)\Lambda(N-n), \qquad \mathfrak{S}(N) = 2\prod_{p>2}\Big(1 - \tfrac{1}{(p-1)^2}\Big) \prod_{\substack{p\mid N \\ p>2}}\Big(1 + \tfrac{1}{p-2}\Big).$$

Write $EH_\mu(Q)$ for the assertion that for every $A > 0$

$$\sum_{q\le Q} \max_{y<N} \max_{(a,q)=1} \Big| \sum_{\substack{n\le y \\ n\equiv a\ (q)}} \Lambda(n)\mu(N-n) - \frac{1}{\varphi(q)} \sum_{n\le y} \Lambda(n)\mu(N-n) \Big| \ll_A \frac{N}{(\log N)^A}. \qquad (1)$$

Huang and Li [2], continuing Pan [8], prove that $EH(N^{\theta}(\log N)^{2A+8})$ together with $EH_{\mu}(N^{1-\theta})$ implies

$$\tilde{r}(N) \;\geq\; \mathfrak{S}(N)\big(1-\mathfrak{A}(N)\big)N + O\Big(\frac{N}{(\log N)^A}\Big), \tag{2}$$

and deduce (their Corollary 1) that, in view of Bombieri–Vinogradov, binary Goldbach for all sufficiently large even integers follows from $EH_{\mu}(N^{\theta'})$ alone for any single $\theta' > 1/2$. Here

$$\mathfrak{A}(N) \;=\; \prod_{\substack{q \nmid N \\ q>2}} \Big(1 - \frac{1}{q(q-1)}\Big) \tag{3}$$

is their equation (7); N is even, so $q > 2$ is implied by $q \nmid N$ and is written here only to match their display. We write $\mathfrak{A}$ rather than their A because A is already the free exponent in (1), and one symbol must not carry two meanings.

This is the sharpest conditional frame we are aware of: one sentence stands between classical technology and Goldbach. This paper asks what that sentence costs on the side where it is consumed.

1.2 Where EH_{μ} is consumed

The hypothesis (1) enters the Huang–Li proof at exactly two places. With α the Vaughan-type parameter of their §3 and $K = (N-1)/\alpha$, these are (their notation; note that the residue class is the *fixed* class $a \equiv N$):

$$E_3(\alpha) = \sum_{\substack{k<K \\ (k,N)=1}} \mu(k)\log k\Big[\sum_{\substack{n<N \\ n\equiv N\ (k)}} \Lambda(n)\mu(N-n) - \frac{1}{\varphi(k)}\sum_{n<N}\Lambda(n)\mu(N-n)\Big], \tag{4}$$

$$E_4(\alpha) = \sum_{\substack{k<K \\ (k,N)=1}} \mu(k)\Big[\sum_{\substack{n<N \\ n\equiv N\ (k)}} \Lambda(n)\mu(N-n)\log(N-n) - \frac{1}{\varphi(k)}\sum_{n<N}\Lambda(n)\mu(N-n)\log(N-n)\Big]. \tag{5}$$

In both cases Huang–Li discard the signs $\mu(k)$ by the triangle inequality on the first line and then appeal to (1) (their Lemma 4 for E_4). The two functionals differ in exactly one respect: E_3 carries the weight $w_k = \log k$ and E_4 carries $w_k = 1$ (the factor $\log(N-n)$ inside E_4 is k-independent and is removed by partial summation). This paper shows that this one difference decides everything.

1.3 Results

For $(k, N) = 1$ and $1 \leq t < N$ put

$$E_{\mu}(t;k) := \sum_{\substack{n\leq t \\ n\equiv N\ (k)}} \Lambda(n)\mu(N-n) - \frac{1}{\varphi(k)}\sum_{n\leq t}\Lambda(n)\mu(N-n), \qquad C(t) := \sum_{n\leq t}\Lambda(n)\mu(N-n), \tag{6}$$

and, for a weight w,

$$T_w(t) \;:=\; \sum_{\substack{k<K \\ (k,N)=1}} \mu(k)\,w(k)\,E_{\mu}(t;k).$$

Thus E_4 is obtained from T_1 and E_3 from $T_{\log}$.

Theorem 1. *Fix $\theta' \in (1/2, 1)$ and put $K = N^{\theta'}$. Then for every $A > 0$,*

$$\sup_{1\le t<N} \big|T_1(t)\big| \;=\; \sup_{1\le t<N} \Big| \sum_{\substack{k<K\\(k,N)=1}} \mu(k)\, E_\mu(t;k) \Big| \;\ll_{A,\theta'}\; \frac{N}{(\log N)^A}.$$

Moreover every ingredient of the proof except Bombieri–Vinogradov contributes only $O\Big(Ne^{-c\sqrt{\log N}}\Big)$: the power of log *in the statement is imposed by Bombieri–Vinogradov alone.*

Theorem 1 is a statement about a *signed* sum in a *fixed* residue class. It is not an instance of, and does not imply, $EH_\mu(N^{\theta'})$, which asks for absolute values and a maximum over residue classes. The mechanism is described in Section 2: after divisor switching one *completes* the divisor sum, and on the complementary range $k \ge K$ the Möbius factor on the long variable squares itself away while the surviving Möbius sits on a variable of size $\le N^{1/2-\delta}$. The completion is not a formality — without it the cofactor is not short and the configuration is the one with no known machine.

Corollary 2. *In the Corollary-1 regime, $E_4(\alpha) \ll_A N(\log N)^{-A}$ for every $A > 0$, unconditionally. Consequently Huang–Li's estimate $S_4(\alpha) = O(N(\log N)^{-A})$ requires no hypothesis, their Lemma 4 is not needed, and the entire EH_μ demand of their argument collapses to the single scalar $E_3(\alpha)$ — together with the term Δ of Section 7 where the published form of (18) is used, which is likewise unconditional there, and which does not arise on the corrected form.*

Theorem 3. *In the Corollary-1 regime one has the unconditional identity*

$$E_3(\alpha) \;=\; \sum_{n<N} \Lambda(n)\Lambda(N-n) \;-\; \mathfrak{S}(N)\Big(N - \sum_{n<N} \Lambda(n)\mu(N-n)\Big) \;+\; O_A\Big(\frac{N}{(\log N)^A}\Big).$$

In particular the bound $E_3(\alpha) \ll_A N(\log N)^{-A}$ — the form of EH_μ that the Huang–Li argument consumes, though not the weakest that suffices; see Proposition 5 — is equivalent *to Huang–Li's equation (22). It therefore yields binary Goldbach for large even N through their Theorem 1; and, granted it, the asymptotic $\tilde r(N) \sim \mathfrak{S}(N)N$ holds if and only if $C(N) = o(N)$.*

Note 4 (three statements, not two). The identity says that the displayed difference *is* $E_3(\alpha)$, so the hypothesis $E_3 \ll_A N(\log N)^{-A}$ and Huang–Li's (22) are the same assertion. What [2] record immediately after their (22) is a different equivalence, between $\tilde r(N) \sim \mathfrak{S}(N)N$ and $C(N) = o(N)$. Chaining the two, $E_3 \ll_A N(\log N)^{-A}$ delivers binary Goldbach — Huang–Li's Theorem 1 supplies $|C(N)| \le \mathfrak{A}(N)N(1 + o(1))$ by the triangle inequality, and $\mathfrak{A}(N) < 1$ — but it does not on its own deliver the asymptotic, which needs $C(N) = o(N)$ in addition. The three statements must be kept apart.

Theorem 3 is the more consequential half. It says that the demand side is closed at the level of *identities* rather than of *estimates*: no choice of θ', of truncation, or of smoothing can evade it, because the root cause is the identity $\mu * \log = \Lambda$ from which Huang–Li start. The weight $\log k$ is the carrier of the Goldbach content, so every divisor switch must hand it back.

Closure at the level of identities is not closure at the level of *strength*, and the next two propositions separate the two. Every condition on E_3 unwinds, through Theorem 3, to a condition on $\tilde r(N)$ and $C(N)$ — that is the identity part, and it is permanent. But Huang–Li consume E_3 two-sidedly and at a saving of every power of log, and Goldbach needs neither.

Proposition 5 (the demand is one-sided, and its threshold is not a log power)**.** *In the Corollary-1 regime, binary Goldbach for all sufficiently large even N follows from*

$$E_3(\alpha) \;>\; -\,\mathfrak{S}(N)\big(1 - \mathfrak{A}(N)\big)N\,\big(1 + o(1)\big), \tag{7}$$

with $\mathfrak{A}(N)$ as in (3). This is strictly weaker than $E_3 \ll_A N(\log N)^{-A}$ in two independent ways: it constrains E_3 from one side only, and its threshold is $\mathfrak{S}(N)(1-\mathfrak{A}(N))N$, which is $\asymp N$ for almost all even N and never smaller than $c\,N/(\log N \log\log N)$ for an absolute constant $c > 0$.

Proposition 6 (the demand needs no saving in $\log N$)**.** *Put*

$$B(N) \;:=\; \sum_{\substack{k<K\\(k,N)=1}} \mu^2(k)\,(\log k)\,\bigl|E_\mu(N;k)\bigr|, \tag{8}$$

the restriction to squarefree k being what the triangle inequality on (4) leaves, since $\mu(k) = 0$ elsewhere. This is sharper than the quantity [2] actually reach before appealing to EH_μ, namely $\log N \sum_{k<K,(k,N)=1} |E_\mu(N;k)|$, in which $\log k$ has been raised to $\log N$; the proposition below holds for either, and a fortiori for theirs. Then binary Goldbach for all sufficiently large even N follows from

$$B(N) \;\le\; (1-\varepsilon)\,\mathfrak{S}(N)\bigl(1-\mathfrak{A}(N)\bigr)N \tag{9}$$

for a single fixed $\varepsilon > 0$. Since $\mathfrak{S}(N)(1-\mathfrak{A}(N)) \asymp 1$ for almost all even N, (9) is a constant-factor *bound on exactly the object EH_μ bounds — absolute values kept, and no saving in $\log N$ whatsoever — where $EH_\mu(N^{\theta'})$ asks for $\ll_A N(\log N)^{-A}$ for every A.*

1.4 What is not claimed

- **No progress toward Goldbach.** Theorem 1 is unconditional but Goldbach-neutral: it removes exactly the half of the demand that carries no Goldbach content. The net progress toward the Goldbach conjecture is zero.
- **No new estimate for $C(N)$.** Theorem 3 places the whole surviving demand on $C(N) = \sum_{n<N} \Lambda(n)\mu(N-n)$; nothing here bounds that quantity.
- **No claim of novelty of mechanism.** The ingredients of Theorem 1 are classical (Section 8). Its difficulty is bookkeeping, and the two places where the bookkeeping bites are isolated in Section 2 and Remark 9.
- **The weakenings do not reopen the design space.** Propositions 5 and 6 lower the required strength but not the required level; the divisor-switch route remains closed over its whole weight space [7].
- **The defect report is not a retraction.** [2]'s Theorem 1 and Corollary 1 stand as stated; what Section 7 supplies is a missing term and its treatment.

2 Notation and the mechanism

p and q always denote primes. φ is Euler's function, τ_3 the 3-fold divisor function, $\mathrm{rad}(u) = \prod_{p|u} p$. Constants $c, c_1, \ldots$ are positive and absolute unless subscripted; implied constants may depend on A and θ'. The letter A is reserved for the free exponent of (1); the local factor of (3) is $\mathfrak{A}$.

The mechanism of Theorem 1 is the following. The residue class in (6) is the fixed class $n \equiv N \pmod k$, i.e.

$$n \equiv N \pmod k \quad\Longleftrightarrow\quad k \mid N-n.$$

So the k-sum is a divisor sum over $u = N - n$, namely the incomplete sum $\sigma_K(u) = \sum_{k|u,\ k<K,\ (k,N)=1} \mu(k)$, and the mechanism has three steps rather than one.

First one **completes** it. By Lemma 7, for squarefree u the complete sum $\sum_{k|u,\,(k,N)=1} \mu(k)$ equals $\mathbf{1}_{\mathrm{rad}(u)|N}$, which forces $u \mid \mathrm{rad}(N)$ and so leaves only $N^{o(1)}$ terms. *Then* the work sits entirely in the complementary sum, over $k \geq K$ — and it is only on that range that writing $u = mk$ gives

$$m \;=\; u/k \;<\; N/K \;=\; N^{1-\theta'} \;\leq\; N^{1/2-\delta}.$$

Finally, for squarefree $u = mk$ the factorisation forces $(m,k) = 1$ and $\mu(u)\mu(k) = \mu(m)\mu^2(k)$: the Möbius factor on the *long* variable k cancels against itself and leaves the nonnegative μ^2, while the surviving Möbius sits on the *short* variable m. That is the classical "Möbius on the short variable plus Bombieri–Vinogradov" configuration.

The completion step is load-bearing and easy to omit. Over the range $k < K$ as it stands the cofactor u/k runs up to u itself, so the bound $m < N^{1-\theta'}$ is simply false there and the surviving Möbius would sit on the *long* variable — the configuration for which no machine is known. Steps 1–3 of Section 3 carry out the argument in that order.

3 Proof of Theorem 1

Fix $A > 0$ and set

$$D_0 := (\log N)^{A+2}, \qquad E_0 := (\log N)^{2A+4}. \tag{10}$$

The truncations must be allowed to depend on A; a fixed choice covers only bounded A.

3.1 Step 0: the subtracted mean term

By (6),

$$T_1(t) \;=\; D(t) \;-\; C(t)\,B(K), \qquad D(t) := \sum_{\substack{k<K\\(k,N)=1}} \mu(k) \sum_{\substack{n\leq t\\ n\equiv N\ (k)}} \Lambda(n)\mu(N-n), \tag{11}$$

where $B(K) = \sum_{k<K,(k,N)=1} \mu(k)/\varphi(k)$. Huang–Li's Lemma 1 (a form of Goldston–Yıldırım [1]) gives $B(K) \ll e^{-c_1\sqrt{\log K}}$, and trivially $|C(t)| \leq \sum_{n<N} \Lambda(n) \ll N$. Hence

$$|C(t)B(K)| \;\ll\; Ne^{-c\sqrt{\log N}} \tag{12}$$

uniformly in t.

3.2 Step 1: divisor switching (an exact identity)

Since $n \equiv N \pmod k$ with $n \leq t < N$ is the same as $u := N - n$ being a multiple of k in $[N-t, N)$,

$$D(t) \;=\; \sum_{N-t\leq u<N} \Lambda(N-u)\,\mu(u)\,\sigma_K(u), \qquad \sigma_K(u) := \sum_{\substack{k|u,\ k<K\\(k,N)=1}} \mu(k). \tag{13}$$

This is a finite rearrangement, hence exact. It is verified independently by a script that accumulates the two sides in genuinely different index orders — the k-side by modular slices, the u-side against a divisor-sum array — and finds them equal to 10^{-16} relative to N (`audit_switch_identity.py`).

3.3 Step 2: the complete divisor sum

Lemma 7. *For squarefree* $u \geq 1$, $\displaystyle\sum_{k|u,\ (k,N)=1} \mu(k) = \mathbf{1}_{\mathrm{rad}(u)|N}$.

Proof. The sum is multiplicative in u with local factor 1 at $p \mid N$ and $1 - 1 = 0$ at $p \nmid N$. Hence it vanishes unless every prime of u divides N. □

Splitting $\sigma_K(u)$ into the complete sum minus the tail $k \geq K$, (13) becomes $D(t) = P(t) - R(t)$ with

$$P(t) = \sum_{\substack{N-t \leq u < N \\ \mathrm{rad}(u) \mid N}} \Lambda(N-u)\mu(u), \qquad R(t) = \sum_{N-t \leq u < N} \Lambda(N-u)\mu(u) \sum_{\substack{k \mid u,\ k \geq K \\ (k,N)=1}} \mu(k). \tag{14}$$

In $P(t)$ the conditions $\mu(u) \neq 0$ and $\mathrm{rad}(u) \mid N$ force $u \mid \mathrm{rad}(N)$, so there are at most $2^{\omega(N)} = N^{o(1)}$ terms, each $\ll \log N$:

$$P(t) \ll N^{o(1)}. \tag{15}$$

3.4 Step 3: the residual, and the degeneracy lemma

Write $u = mk$ with $k \geq K$, so $m = u/k < N/K =: M$. For squarefree u the factorisation forces $(m, k) = 1$ and $\mu(u)\mu(k) = \mu(m)\mu^2(k)$, so

$$R(t) = \sum_{m<M} \mu(m) \sum_{\substack{k \geq K,\ (k,m)=1,\ (k,N)=1 \\ N-t \leq mk < N}} \mu^2(k)\, \Lambda(N - mk). \tag{16}$$

Lemma 8 (degeneracy)**.** *The contribution to (16) of the terms with $(k, N) > 1$ is $O(N^{o(1)})$. The same bound holds for the terms in which the modulus constructed in Step 4 is not coprime to N.*

Proof. Suppose $p \mid (k, N)$ and $\Lambda(N - mk) \neq 0$, say $N - mk = q^\ell$. Since $p \mid k$ and $p \mid N$ we get $p \mid N - mk$, hence $q = p$ and $n := N - mk = p^\ell$ with $p \mid N$. The number of such $n < N$ is $\leq \sum_{p \mid N} \log N/\log p \ll (\log N)^2$; for each, the pairs (m, k) with $mk = N - n$ number $\leq \tau(N - n) \ll N^{o(1)}$, and each term is $\ll \log N$. □

By Lemma 8 we may drop the condition $(k, N) = 1$ from (16) at a cost of $O(N^{o(1)})$. This is essential: expanding $\mathbf{1}_{(k,N)=1}$ by Möbius over $e \mid N$ would multiply the number of Bombieri–Vinogradov calls by $2^{\omega(N)} = N^{o(1)}$ and destroy the budget.

3.5 Step 4: unfolding μ^2 and the coprimality

Insert $\mu^2(k) = \sum_{d^2 \mid k} \mu(d)$ and $\mathbf{1}_{(k,m)=1} = \sum_{e \mid (k,m)} \mu(e)$ and truncate at D_0, E_0 of (10).

Tail $d > D_0$. Such terms have $d^2 \mid k$, hence $n = N - mk \equiv N \pmod{md^2}$; the number of $n \leq N$ in that progression is $\ll N/(md^2) + 1$ and each $\Lambda \ll \log N$. Summing,

$$\ll \log N \sum_{m<M} \sum_{d>D_0} \left(\frac{N}{md^2} + 1\right) \ll \frac{N(\log N)^2}{D_0} + \sqrt{NM} \ll \frac{N}{(\log N)^A} + N^{1-\theta'/2}.$$

Tail $e > E_0$. Such terms have $e \mid k$ and $e \mid m$, hence $n \equiv N \pmod{me}$, and

$$\ll \log N \sum_{m<M} \sum_{\substack{e \mid m \\ e > E_0}} \left(\frac{N}{me} + 1\right) \ll \frac{N \log N}{E_0} \sum_{m<M} \frac{\tau(m)}{m} + M \ll \frac{N(\log N)^3}{E_0} + M.$$

Both are $\ll N(\log N)^{-A}$. In the remaining range put $L := \mathrm{lcm}(d^2, e)$ and $q := mL$; the conditions $d^2 \mid k$, $e \mid k$ and $n = N - mk$ give $n \equiv N \pmod q$, and $mk < N$, $mk \geq \max(N - t, mK)$ give $1 \leq n \leq \mathcal{T}_m(t)$ where

$$\mathcal{T}_m(t) := \min\bigl(t,\ N - mK\bigr). \tag{17}$$

Therefore, with $\psi(y;q,a) = \sum_{n\le y,\, n\equiv a\,(q)} \Lambda(n)$,

$$R(t) = \sum_{m<M} \mu(m) \sum_{d\le D_0} \mu(d) \sum_{\substack{e\mid m\\ e\le E_0}} \mu(e)\, \psi\big(\mathcal{T}_m(t);\, q,\, N\big) \;+\; O\bigg(\frac{N}{(\log N)^A}\bigg). \tag{18}$$

By Lemma 8 we may further restrict to $(q,N)=1$: if $g=(q,N)>1$ then $g\mid n$ forces n to be a power of a prime dividing N, and the total over the $\ll MD_0E_0\tau$-many triples is $\ll N^{1-\theta'+o(1)}$.

Note 9 (the false-positive trap). Restricting the *main terms* to classes with $(q,N)=1$ is not cosmetic. Assigning a main term $\mathcal{T}_m/\varphi(q)$ to a degenerate class shifts the density of the whole computation by the factor $N/\varphi(N)$. Carried into the $\log k$ branch of Section 5, that error produces an apparent *refutation* of EH_μ. It is recorded here because it is the most plausible false positive in this circle of ideas.

Note the size of the modulus: $q = m\,\mathrm{lcm}(d^2,e) \le MD_0^2E_0$, so by (10)

$$q \;\le\; N^{1-\theta'}(\log N)^{4A+8} \;=\; N^{1/2-\delta}(\log N)^{4A+8} \;\le\; N^{1/2-\delta/2} \tag{19}$$

for N large. This is the level at which Bombieri–Vinogradov is applied.

3.6 Step 5: Bombieri–Vinogradov

Lemma 10 (τ-weighted Bombieri–Vinogradov)**.** *For all $A,B>0$ there is $C=C(A,B)$ such that for $Q\le N^{1/2}(\log N)^{-C}$,*

$$\sum_{q\le Q} \tau_3(q)^B \max_{y\le N} \max_{(a,q)=1} \bigg|\psi(y;q,a) - \frac{y}{\varphi(q)}\bigg| \;\ll_{A,B}\; \frac{N}{(\log N)^A}.$$

Proof. Write $\mathcal{E}_q$ for the inner maximum. By Cauchy–Schwarz,

$$\sum_q \tau_3^B \mathcal{E}_q \;\le\; \Big(\sum_q \tau_3^{2B}\mathcal{E}_q\Big)^{1/2}\Big(\sum_q \mathcal{E}_q\Big)^{1/2}.$$

For the first factor use the trivial bound $\mathcal{E}_q \ll (N/\varphi(q))\log N$, giving $\ll N(\log N)^{C_1(B)}$; for the second use Bombieri–Vinogradov [6] with exponent $2A + C_1(B)$. $\square$

A modulus q arises in (18) from at most $\sum_{m\mid q} \tau(q/m)^2 \le \tau(q)^3 \le \tau_3(q)^3$ triples (m,d,e): given $m\mid q$ there are at most $\tau(q/m)$ choices of d with $d^2\mid q/m$ and at most $\tau(q/m)$ of e with $\mathrm{lcm}(d^2,e)=q/m$. Writing $\psi(\mathcal{T}_m;q,N) = \mathcal{T}_m/\varphi(q) + \mathcal{E}$ and applying Lemma 10 with $B=3$ and (19),

$$R(t) \;=\; \mathrm{MT}(t) \;+\; O\bigg(\frac{N}{(\log N)^A}\bigg), \qquad \mathrm{MT}(t) = \sum_{\substack{m<M\\ (m,N)=1}} \mu(m)\, \mathcal{T}_m(t)\, c_{D_0,E_0}(m), \tag{20}$$

where $c_{D_0,E_0}(m) = \sum_{d\le D_0} \sum_{e\mid m,\, e\le E_0} \mu(d)\mu(e)\mathbf{1}_{(d,N)=1}/\varphi(m\,\mathrm{lcm}(d^2,e))$.

3.7 Step 6: the density

Lemma 11. *Let m be squarefree with $(m,N)=1$ and put*

$$c(m) := \sum_{d\ge 1} \sum_{e\mid m} \frac{\mu(d)\mu(e)\mathbf{1}_{(d,N)=1}}{\varphi(m\,\mathrm{lcm}(d^2,e))}.$$

Then $c(m) = \mathfrak{A}(N)\lambda(m)/m$ with $\mathfrak{A}(N)$ as in (3) and $\lambda(m) = \prod_{p\mid m}\big(1-\frac{1}{p(p-1)}\big)^{-1}$. Moreover $c_{D_0,E_0}(m) = c(m) + O(1/(mD_0))$.

Proof. The sum is multiplicative. If $p \nmid m$ then $p \nmid e$ and the local factor is $1 - 1/\varphi(p^2) = 1 - 1/(p(p-1))$ for $p \nmid N$, and 1 for $p \mid N$. If $p \mid m$ (so $p \nmid N$) the p-part of $m\,\mathrm{lcm}(d^2, e)$ is $p^{1+\max(2v_p(d), v_p(e))}$, and the four choices $(v_p(d), v_p(e)) \in \{0,1\}^2$ contribute

$$\frac{1}{p-1} - \frac{1}{p(p-1)} - \frac{1}{p^2(p-1)} + \frac{1}{p^2(p-1)} = \frac{1}{p}.$$

Hence $c(m) = \prod_{p\mid m} p^{-1} \cdot \prod_{p\nmid m,\ p\nmid N}(1 - \frac{1}{p(p-1)}) = m^{-1}\mathfrak{A}(N)\lambda(m)$. The truncation error is the tail $\sum_{d>D_0} 1/\varphi(md^2) \ll 1/(mD_0)$ and similarly for e. $\square$

The load-bearing line is the local factor. Equivalently, for squarefree m,

$$\sum_{g\mid m} \frac{\mu(g)}{\varphi(m/g)\, g\, \varphi(g)} = \frac{1}{m} :$$

the local factor is exactly p^{-1}, so the density exponent is exactly 1 and $1/\zeta$ occurs to the *first* power in Lemma 12 below. A non-integral exponent would give, by Selberg–Delange, only $(\log x)^{-c}$ for a fixed c in Lemma 12, and Theorem 1 would be false. The identity is verified in exact rational arithmetic for all squarefree $m < 400$, with zero mismatches (`audit_density_identity.py`).

Lemma 12. *Let $f(m) := \mu(m)\lambda(m)\mathbf{1}_{(m,N)=1}$ and $G(x) := \sum_{m\le x} f(m)/m$. Then*

$$G(x) \ll \frac{N}{\varphi(N)}\, e^{-c\sqrt{\log x}} + x^{-1/4}e^{C\sqrt{\log N}} \qquad (x \ge 2).$$

Proof. Write $F(s) = \sum_m f(m)m^{-s} = \prod_{p\nmid N}(1 - \lambda(p)p^{-s})$ and $F(s) = \zeta(s)^{-1}H(s)$, so $f = \mu * h$ with h multiplicative, $h(p^j) = 1$ for $p \mid N$ and $h(p^j) = 1 - \lambda(p) = O(p^{-2})$ for $p \nmid N$. Then $G(x) = \sum_{b\le x} \frac{h(b)}{b} \sum_{a\le x/b} \frac{\mu(a)}{a}$. For $b \le \sqrt{x}$ use $\sum_{a\le y} \mu(a)/a \ll e^{-c\sqrt{\log y}}$ (the classical zero-free region) together with $\sum_b |h(b)|/b \ll \prod_{p\mid N}(1 - 1/p)^{-1} \ll N/\varphi(N)$. For $b > \sqrt{x}$ bound $\sum_{b>y} |h(b)|/b \le y^{-1/2} \sum_b |h(b)|b^{-1/2} \ll y^{-1/2} \prod_{p\mid N}(1 - p^{-1/2})^{-1} \ll y^{-1/2}e^{C\sqrt{\log N}}$. $\square$

Since $M = N^{1-\theta'}$ is a fixed power of N, the second term of Lemma 12 is negligible at $x \asymp M$, and $N/\varphi(N) \ll \log\log N$; so $G(x) \ll e^{-c'\sqrt{\log x}}$ in the relevant range.

3.8 Step 7: the main term dies, uniformly in t

Proposition 13. $\sup\limits_{1\le t<N} |\mathrm{MT}(t)| \ll Ne^{-c\sqrt{\log N}}$.

Proof. By Lemma 11, $\mathrm{MT}(t) = \mathfrak{A}(N) \sum_{m<M} \frac{f(m)}{m} \mathcal{T}_m(t) + O(M\log N/D_0)$ with $\mathcal{T}_m(t)$ as in (17). Put $m_0 := (N-t)/K$, so $\mathcal{T}_m(t) = t$ for $m \le m_0$ and $\mathcal{T}_m(t) = N - mK$ for $m > m_0$; in particular $x \mapsto T_x(t)$ is non-increasing, is constant on $[1, m_0]$, has derivative $-K$ on (m_0, M), and $T_M(t) = 0$. Abel summation against G of Lemma 12 gives

$$\sum_{m<M} \frac{f(m)}{m}\mathcal{T}_m(t) = G(M)\,T_M(t) + K\int_{m_0}^{M} G(x)\,dx = K\int_{m_0}^{M} G(x)\,dx.$$

Hence, uniformly in t,

$$\Big|\sum_{m<M} \frac{f(m)}{m}\mathcal{T}_m(t)\Big| \le K\int_1^M |G(x)|\,dx \ll K\int_1^M e^{-c\sqrt{\log x}}dx \ll KMe^{-c'\sqrt{\log M}} \ll Ne^{-c''\sqrt{\log N}},$$

using $\log M \asymp \log N$. The truncation error is $\ll M\log N/D_0 = o(N(\log N)^{-A})$. $\square$

The cancellation is genuine and global: the term $m = 1$ alone contributes $\mathcal{T}_1(t) \asymp N$ — the endpoint of (17), not the weighted sum of Theorem 1 — so the smallness of MT is entirely due to the cancellation in $\sum f(m)/m$.

3.9 Conclusion

Combining (11), (12), (15), (20) and Proposition 13:

$$T_1(t) = \underbrace{P(t)}_{N^{o(1)}} - \underbrace{\mathrm{MT}(t)}_{\ll Ne^{-c\sqrt{\log N}}} - \underbrace{O\big(N(\log N)^{-A}\big)}_{\mathrm{BV}} - \underbrace{C(t)B(K)}_{\ll Ne^{-c\sqrt{\log N}}},$$

uniformly in $t < N$. This proves Theorem 1, and exhibits Bombieri–Vinogradov as the only ingredient that does not give an exponential saving. The bound cannot be improved to $Ne^{-c\sqrt{\log N}}$ by this argument: the Bombieri–Vinogradov input yields $N(\log N)^{-A}$ for each fixed A and no more, because the Siegel–Walfisz range in its proof is $q \leq (\log N)^B$ with B fixed. □

4 Proof of Corollary 2

Fix k and set $a_n := \Lambda(n)\mu(N-n)\big(\mathbf{1}_{n\equiv N\,(k)} - 1/\varphi(k)\big)$, so that $E_\mu(t;k) = \sum_{n\leq t} a_n$ and the bracket in (5) is $\sum_{n<N} a_n \log(N-n)$. Since $\log(N-n)$ vanishes at $n = N-1$ and has derivative $-1/(N-t)$, Abel summation gives

$$\sum_{n<N} a_n \log(N-n) \;=\; \int_1^{N-1} \frac{E_\mu(t;k)}{N-t}\,dt.$$

The weight $\log(N-n)$ does not depend on k, so multiplying by $\mu(k)$ and summing over $k < K$ with $(k,N) = 1$ (a finite sum) gives the exact identity

$$E_4(\alpha) \;=\; \int_1^{N-1} \frac{T_1(t)}{N-t}\,dt.$$

Hence $|E_4(\alpha)| \leq \big(\sup_{t<N} |T_1(t)|\big)\log N \ll_A N(\log N)^{1-A}$ by Theorem 1. As A is arbitrary this is Corollary 2. Huang–Li's Lemma 4 is invoked only to bound E_4; with the above it is not needed, and the only remaining appearance of EH_μ in their §3 is through $E_3(\alpha)$. □

5 Proof of Theorem 3: the permanent closure

Take the weight $w_k = \log k$, i.e. the functional E_3 of (4). Steps 0–1 are unchanged, and the complete divisor sum of Lemma 7 is replaced by the following.

Lemma 14. *For squarefree u, write $u = u_N u'$ where u_N is the largest divisor of u composed of primes dividing N. Then $\sum_{k|u,\ (k,N)=1} \mu(k)\log k = -\Lambda(u')$.*

Proof. The sum is over $k \mid u'$ and equals $-\Lambda(u')$ by $\mu * \log = \Lambda$. □

Hence the complete piece of E_3 is

$$-\sum_{1\leq u<N} \Lambda(N-u)\,\mu(u)\,\Lambda(u') \;=\; -\sum_{u<N,\ (u,N)=1} \Lambda(N-u)\mu(u)\Lambda(u) + O(N^{o(1)}\log^2 N).$$

Now $\mu(u)\Lambda(u)$ is supported on primes $u = p$, where it equals $-\log p$; prime powers $u = p^\ell$, $\ell \geq 2$, have $\mu(u) = 0$. So the complete piece equals

$$\sum_{p<N} \Lambda(N-p)\log p \;+\; O\big(N^{o(1)}\log^2 N\big) \;=\; \sum_{n<N} \Lambda(n)\Lambda(N-n) + O\big(N^{1/2+\varepsilon}\big), \tag{21}$$

the binary Goldbach sum itself. The door is free: divisor switching costs nothing, and immediately behind it stands the object one was trying to avoid.

Two further terms have to be evaluated, and neither vanishes.

(i) The residual main term. Repeating Steps 3–6 with the extra factor $\log k = \log((N-n)/m)$ produces the main term $\mathfrak{A}(N)\sum_{m<M}\mu(m)\lambda(m)m^{-1}\int_{mK}^{N}\log(v/m)\,dv$. Substituting $v = mt$ evaluates the integral as $N\log(N/m) - N - mK\log K + mK$, of which only the $-N\log m$ piece survives the cancellation: by Lemma 12 the coefficients $\sum f(m)/m$ and $\sum f(m)$ both die, and what is left is $-\mathfrak{A}(N)\,N\,\widetilde{G}(1)$ with $\widetilde{G}(1) = \lim_x \sum_{m\le x}\mu(m)\lambda(m)\mathbf{1}_{(m,N)=1}\log m/m$. The constant is fixed by

$$\mathfrak{A}(N)\,\widetilde{G}(1) \;=\; -\mathfrak{S}(N), \tag{22}$$

so that the residual main term is $+\mathfrak{S}(N)\,N + O(N(\log N)^{-A})$.

The sign in (22) is forced, and it is the whole identity: with $F(s) = \sum_m f(m)m^{-s} = \zeta(s)^{-1}H(s)$ as in Lemma 12 one has $\widetilde{G}(1) = -F'(1) = -H(1) < 0$, while $\mathfrak{A}(N)H(1) = \mathfrak{S}(N)$ identically — the local factors pair as

$$\Big(1 - \tfrac{1}{p(p-1)}\Big)\Big(1 - \tfrac{1}{(p^2-p-1)(p-1)}\Big) = 1 - \tfrac{1}{(p-1)^2}$$

at $p \nmid N$, and as $p/(p-1)$ at $p \mid N$. Numerically, $\mathfrak{A}(N)\widetilde{G}(x) = -1.760250$ at $x = 4\cdot 10^6$ against $\mathfrak{S}(N) = 1.760432$ (`audit_E3_constant.py`).

(ii) The subtracted mean term. It carries the factor

$$B_{\log}(K) := \sum_{\substack{k<K\\(k,N)=1}} \frac{\mu(k)\log k}{\varphi(k)} \;=\; -\mathfrak{S}(N) + O\big(e^{-c\sqrt{\log K}}\log K\big),$$

which is exactly Huang–Li's Lemma 1: subtracting $\sum_{k\le K}\mu(k)\varphi(k)^{-1}\log(K/k) = \mathfrak{S}(N) + O(e^{-c_1\sqrt{\log K}})$ from $\log K\cdot\sum_{k\le K}\mu(k)/\varphi(k) = O(\log K\, e^{-c_1\sqrt{\log K}})$ gives the claim. Independently: with $f(s) = \prod_{p\nmid N}\big(1 - \frac{p^{-s}}{p-1}\big)$ one has $f(s) = \zeta(s+1)^{-1}h(s)$ with $h(0) = \mathfrak{S}(N)$, so $B_{\log}(\infty) = -f'(0) = -\mathfrak{S}(N)$. Since the mean term is $-C(N)B_{\log}(K)$ with $C(N) = \sum_{n<N}\Lambda(n)\mu(N-n)$, it contributes $+\mathfrak{S}(N)\,C(N)$.

Collecting (21), (i) and (ii):

$$E_3(\alpha) = \sum_{n<N}\Lambda(n)\Lambda(N-n) - \mathfrak{S}(N)N + \mathfrak{S}(N)C(N) + O_A\bigg(\frac{N}{(\log N)^A}\bigg),$$

which is the assertion of Theorem 3. Since $\mathfrak{S}(N) \asymp 1$, the bound $E_3(\alpha) \ll_A N(\log N)^{-A}$ is therefore equivalent to

$$\tilde{r}(N) = \mathfrak{S}(N)\big(N - C(N)\big) + O_A\big(N(\log N)^{-A}\big),$$

which is Huang–Li's equation (22). Given that, $\tilde{r}(N)$ is positive for large even N — binary Goldbach — because $|C(N)| \le \mathfrak{A}(N)N(1+o(1))$ with $\mathfrak{A}(N) < 1$; and $\tilde{r}(N) \sim \mathfrak{S}(N)N$ holds if and only if $C(N) = o(N)$, which is not implied. □

Note 15 (the identity is not accessible to computation)*.* The finite-N residual $R(N,\theta') = \big|E_3(N;\theta') - (\tilde{r}(N) - \mathfrak{S}(N)(N - C(N)))\big|/N$ does not become small in any accessible range, and it cannot be made small by either free parameter. Over $N = 2\cdot 10^5, 4\cdot 10^5, 8\cdot 10^5$ at $\theta' = 0.56$ it reads 0.4558, 0.3729, 0.3108, while the right-hand side it is compared against is $\asymp 10^{-3}N$. Swept over $\theta' \in [0.51, 0.95]$, R is essentially monotone *increasing* — at $N = 8\cdot 10^5$ from 0.170167 at $\theta' = 0.51$ to 4.991178 at $\theta' = 0.90$ — because the finite-N error is dominated by the main-term cancellation over $m < M = N^{1-\theta'}$, which $\theta' \to 1$ destroys. The best θ' is the smallest admissible one, just above 1/2. Increasing N points the wrong way as well: at the best θ' the ratio of error to signal reads 15.19, 18.38, 26.84 at the three N above, because $C(N) = o(N)$ being true kills the right-hand side fast while the finite-N error dies only at a slow power of log. So Theorem 3 is a statement no computation confirms, at any N; the proof is the only access to it (`lab_theta_sweep.py`).

Note 16. The mechanism is structural, not accidental. The identity $\mu * \log = \Lambda$ is the starting point of the Huang–Li argument (their (10)); it is therefore also what any divisor switch inside their $\log k$-weighted functional must return. No choice of θ', truncation, or smoothing can circumvent an identity. What the identity does not fix is the *strength* at which E_3 must be bounded, and that is Section 6.

6 How strong the surviving demand really is

Proof of Proposition 5. By Theorem 3, $\tilde r(N) = \mathfrak{S}(N)(N - C(N)) + E_3(\alpha) + O_A(N(\log N)^{-A})$. The triangle inequality gives $|C(N)| \le U(N) := \sum_{n<N} \Lambda(n)\mu^2(N-n)$, and $U(N) = \mathfrak{A}(N)N(1+o(1))$ — proved in [2, §3.3] itself by Bombieri–Vinogradov, with error $O(N(\log N)^{-A-1})$, and classically by Mirsky's theorem on squarefree shifted primes [5] together with partial summation. Hence $\mathfrak{S}(N)(N - C(N)) \ge \mathfrak{S}(N)(1 - \mathfrak{A}(N))N(1 + o(1))$, and $\mathfrak{A}(N) < 1$ for every N, so the right-hand side is positive. Substituting and using that the O_A term may be taken below any fixed power of $N/\log N$ gives $\tilde r(N) > 0$ under (7).

For the size of the threshold, $\mathfrak{A}(N) = \prod_{q\nmid N}(1 - \frac{1}{q(q-1)})$ is largest when N carries the most small primes, so $1-\mathfrak{A}(N)$ is smallest at primorials, where $1-\mathfrak{A}(N) \asymp \sum_{q>p} \frac{1}{q(q-1)} \asymp 1/(p\log p)$ with $p \asymp \log N$; and $\mathfrak{S}(N) \gg 1$. $\square$

Proof of Proposition 6. Immediate from Theorem 3, $|E_3(\alpha)| \le B(N)$, and $|C(N)| \le \mathfrak{A}(N)N(1+o(1))$ as above: $\tilde r(N) \ge \mathfrak{S}(N)(1-\mathfrak{A}(N))N(1+o(1)) - B(N) + O_A(N(\log N)^{-A})$. $\square$

Measurement 17 (the threshold, and the margin)**.** Over even $N \le 1.6 \cdot 10^7$ the threshold constant $\mathfrak{S}(N)(1 - \mathfrak{A}(N))$ has median 0.333459 and 0.1-percentile 0.119639; its minimum is 0.060890, attained at $N = 9\,699\,690 = 2 \cdot 3 \cdots 19$, the largest primorial in range, and the ten smallest values are all primorial-like. Multiplied by $\log N \log\log N$ the minimum over $N \ge 10^5$ is 2.482019, attained at $N = 510\,510$; over $N \ge 10^3$ it is 1.737799 at $N = 2310$; over $N \ge 16$ it is 0.810202 at $N = 30$. Every argmin is a primorial, which is the mechanism of Proposition 5 made visible; the constant c there is therefore stated as a constant and not as 1. The step that produces the margin is exact to five decimals: $U(N)/(\mathfrak{A}(N)N)$ has mean 0.999996 with standard deviation 0.000170 over the top octave. Checked against the truth, no even $N \in [10^5, 1.6 \cdot 10^7]$ has $\tilde r(N) < \mathfrak{S}(N)(1 - \mathfrak{A}(N))N$; the slack between them has minimum 3.7038, median 4.2902 and maximum 95.0889, the maximum at $N = 9\,699\,690$ — so the margin is thinnest exactly where the Goldbach count is largest (`lab_onesided_margin.py`).

Measurement 18 (what the weakening relocates)**.** At $\theta' = 0.56$ and $N = 2 \cdot 10^5$ through $3.2 \cdot 10^6$, the quantity $B(N)$ of (8) satisfies

$$B(N)/N \;=\; 0.8086,\; 0.7395,\; 0.7303,\; 0.6547,\; 0.5916$$

against a threshold of $0.3745N$, so the ratio $B(N)/(\mathfrak{S}(N)(1-\mathfrak{A}(N))N)$ falls 2.159, 1.975, 1.950, 1.748, 1.580. The object Proposition 6 constrains is therefore already within a factor 1.6 of the threshold at the top of the accessible range, and falling; but a constant-factor bound valid for *all* large N is what is needed, and no computation supplies that (`lab_onesided_demand.py`).

Note 19 (the threshold is not a constant, and this sweep held it fixed). The 0.3745 above is $\mathfrak{S}(N)(1 - \mathfrak{A}(N))$, and both factors depend on which primes divide N. The N of the sweep are $2 \cdot 10^5 \cdot 2^j$ with $2 \cdot 10^5 = 2^6 \cdot 5^5$, so all of them have odd radical 5 and *one* threshold: the sweep moves the size of N by a factor 128 and its arithmetic not at all.

Changing the arithmetic changes the verdict. At seven N in $[1.40 \cdot 10^6, 1.63 \cdot 10^6]$ chosen for their odd radicals the threshold runs from 0.073312 to 0.374487 — a factor of five, with the swept family sitting at the *maximum* rather than above the median 0.270682 — while $|E_3|/N$ moves the other way, from 0.1289 to 2.7763. The two compound: (7) holds at three of the seven and fails at four. The mechanism is visible: when N carries several small odd primes, k and m must both avoid them, so the surviving m are almost all prime and $\mu(m) = -1$ dominates. At $N = 2 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17$ all 513 terms of E_3 are negative and $|E_3| = B(N)$ exactly — no cancellation across k at all, against 442 of 562 at the swept N.

The damage is not where it looks. Proposition 5 reaches its threshold through $|C(N)| \leq \mathfrak{A}(N)N$, and that bound is slack: $|C|/(\mathfrak{A}N)$ measures 0.0012 to 0.0208 across the seven. With the measured wall in its place, Theorem 3 asks for $|E_3|/N < \mathfrak{S}(N)(1 - |C(N)|/N)$, and *all seven* satisfy it (2.7763 against 5.3505 at the worst). The arithmetic dependence of the verdict therefore lives entirely in the slack of $|C| \leq \mathfrak{A}N$ and not in E_3. This is a measurement, not a proof — $C(N) = o(N)$ is the open problem and nothing here supplies it (`audit_threshold_arithmetic.py`).

Proposition 5 does not reopen the divisor switch. The no-go of [7] costs a factor $\exp(c_1\sqrt{\frac{1}{2}\log N})$, which exceeds $\log N \log\log N$ as it exceeds every power of $\log N$; so the weaker threshold is still out of reach of that design space.

7 The correction Δ to equation (18)

The published form of equation (18) drops a range. What follows records the defect and one way of closing it. Neither is claimed as new: the observation is S. Zheleznov's, and the authors have since corrected the manuscript by a different route, on which the dropped range never appears. What is set down here is kept because the mechanism it uses is the one Theorem 1 rests on, and because Proposition 20 below shows the two routes meet.

The scope of the report has to be stated: we have worked from the arXiv version, and have checked the passage below in **both** v1 (May 2020) and v2 (August 2022), where it appears identically. We have not seen the version published in the Springer volume and make no claim about it.

Huang–Li define

$$S_2(\alpha) = \sum_{n<N} \Lambda(n)\, \mu^2(N-n)\, \tilde{\Lambda}_\alpha(N-n), \qquad \tilde{\Lambda}_\alpha(u) = \sum_{d|u,\ d>\alpha} \mu(d) \log(1/d),$$

and substitute $k = u/d$, so that the constraint $d > \alpha$ becomes

$$k \ < \ \frac{N-n}{\alpha}, \tag{23}$$

an *n-dependent* bound. Their displayed equation (18) reads

$$S_2(\alpha) \ = \sum_{k<\frac{N-1}{\alpha}} \mu(k) \sum_{\substack{n<N \\ n\equiv N\ (k)}} \Lambda(n)\mu(N-n) \log\Big(\frac{k}{N-n}\Big),$$

in which (23) has been replaced by the n-free bound $k < (N-1)/\alpha$. The two differ. Writing $N - n = mk$, the condition (23) is $m > \alpha$, so the right-hand side of (18) contains, in addition to $S_2(\alpha)$, exactly the terms with $m \leq \alpha$. On those terms $\mu(k)\mu(N-n) = \mu(m)\mu^2(k)\mathbf{1}_{(k,m)=1}$ and $\log(k/(N-n)) = -\log m$, so

$$S_2(\alpha) \ = \ \text{RHS of (18)} \ + \ \Delta, \qquad \Delta \ = \sum_{2\leq m\leq\alpha} \mu(m) \log m \sum_{\substack{k<(N-1)/\alpha \\ (k,m)=1}} \mu^2(k)\, \Lambda(N-mk). \tag{24}$$

(The term $m = 1$ drops out since $\log 1 = 0$.) The trivial bound is $\Delta \ll N(\log N)^2$, which exceeds the target $O(N(\log N)^{-A})$, so Δ is not negligible and needs its own treatment.

It does close, by the machinery already used above and under hypotheses Huang–Li already assume. Indeed Δ has exactly the shape of the residual (16): the Möbius factor sits on the *short* variable $m \le \alpha$, and the long variable k carries only $\mu^2 \ge 0$. Since $\mu(m) \ne 0$ on the range of summation, the two conditions on k collapse into one,

$$\mu^2(k)\,\mathbf{1}_{(k,m)=1} \;=\; \mu^2(mk), \tag{25}$$

and expanding $\mu^2(mk) = \sum_{b^2|mk} \mu(b)$ with the truncation $b \le B := (\log N)^{A+4}$ — the same one Huang–Li use in their §3.1 — reduces Δ to sums of Λ over progressions to moduli $[m, b^2] \le \alpha B^2 = \alpha(\log N)^{2A+8}$, plus a main term $\mathfrak{A}(N) \sum_{m\le\alpha} \mu(m)\lambda(m)(\log m)\,\mathcal{T}_m/m$ which is $O(Ne^{-c\sqrt{\log N}})$ by Lemma 12 and partial summation. The level is then exactly the one their hypothesis supplies, with no enlargement of A. Hence:

- In the Corollary-1 regime, $\alpha = N^{1-\theta'} < N^{1/2}$ and Bombieri–Vinogradov closes Δ *unconditionally.*
- In the general Theorem-1 regime, $\alpha \asymp N^{\theta}$ and Δ closes under the hypothesis $EH(N^{\theta}(\log N)^{2A+8})$, which is assumed there.

So [2]'s Theorem 1 and their Corollary 1 stand as stated. In the published text the treatment of Δ is missing; in the corrected text Δ does not arise, because the truncation is not moved off the inner variable in the first place.

7.1 The two formulations differ by Δ

The corrected argument keeps the moving cut: with $Y_k = \lceil N - \alpha k\rceil - 1$ the inner sum stops at Y_k, the terms with $m \le \alpha$ never enter, and (24) has nothing to correct. Every statement above was proved against the fixed cut, so the question is whether it survives the move. It does, and the term that carries the difference is Δ itself.

Write $J = T_1(N-1)$ for the fixed-cut object at its extreme endpoint and $T_1^* = \sum_{k<K,(k,N)=1} \mu(k) E_\mu(Y_k; k)$ for the moving-cut one. The difference is carried by the n the moving cut drops, and there

$$n > Y_k \iff N - n \le \alpha k \iff m \le \alpha, \qquad m = \frac{N-n}{k},$$

so on exactly those terms $\mu(k)\mu(N-n) = \mu(k)\mu(mk)$, which vanishes unless mk is squarefree and equals $\mu(m)\mu^2(k)$ otherwise: the Möbius factor on the *short* variable, the long one carrying only $\mu^2 \ge 0$. That is (24)'s shape with weight 1 in place of $\log m$.

Proposition 20 (the bound survives the corrected cut)**.** *Fix $\theta' \in (1/2, 1)$, put $\alpha = N^{1-\theta'}$ and $K = (N-1)/\alpha$. Then for every $A > 0$, $T_1^* \ll_{A,\theta'} N(\log N)^{-A}$, unconditionally.*

Proof. Exchanging as above, $J - T_1^* = \Xi_{\text{main}} - \Xi_{\text{mean}}$ with

$$\Xi_{\text{main}} = \sum_{m\le\alpha} \mu(m) \sum_{\substack{k<K\\(k,mN)=1}} \mu^2(k)\,\Lambda(N - mk), \qquad \Xi_{\text{mean}} = \sum_{\substack{k<K\\(k,N)=1}} \frac{\mu(k)}{\varphi(k)}\big(C(N-1) - C(Y_k)\big),$$

the constraint $n \ge 1$ being automatic since $m \le \alpha$ and $k < K$ force $mk < N-1$.

$J \ll_{A,\theta'} N(\log N)^{-A}$ is Theorem 1 at $t = N-1$, which lies in the range of its supremum; this is the only place $\theta' > 1/2$ is used.

In Ξ_{main} the term $m = 1$ is $\sum_{k<K,(k,N)=1} \mu^2(k)\Lambda(N-k) \le \sum_{N-K<j<N} \Lambda(j) \ll K \log N = N^{\theta'} \log N$, which is $\ll_A N(\log N)^{-A}$ because $\theta' < 1$. The terms $2 \le m \le \alpha$ are (24) with $\log m$ replaced by 1 and close by the argument above verbatim, the main term dying by Lemma 12 applied directly — the weight being 1, no partial summation is needed to remove a $\log m$.

For Ξ_{mean}, since $n > Y_k$ iff $k > R_n := \lceil (N-n)/\alpha \rceil - 1$, exchanging gives $\Xi_{\mathrm{mean}} = \sum_{n<N} a_n(\rho_N(K) - \rho_N(R_n+1))$ with ρ_N as in [7, Lem. 14] and $a_n = \Lambda(n)\mu(N-n)$. Split at $N-n = H$ with $H := 2N^{1-\theta'/2}$. On $N-n \le H$ use $|\rho_N| \ll 1$ and $\sum_{N-H\le n<N} |a_n| \ll H \log N$, which is $\ll_A N(\log N)^{-A}$ since $1-\theta'/2 < 1$. On $N-n > H$ one has $R_n \ge N^{\theta'/2}$, so both arguments of ρ_N meet its hypothesis against $n = N$ and both terms are $\ll e^{-c\sqrt{\theta' \log N}}$; with $\sum_{n<N} |a_n| \ll N$ this is $\ll_A N(\log N)^{-A}$ as well. □

Note 21 (what the proposition does not cover). It is stated in the Corollary-1 regime only. In the general Theorem-1 regime $\alpha \asymp N^{\theta}$ the outer truncation is $K \asymp N^{1-\theta}$, which drops below $N^{1/2}$ as soon as $\theta > 1/2$, and Theorem 1 is stated for $K = N^{\theta'}$ with $\theta' \in (1/2, 1)$. The exchange and the estimate of Ξ_{main} are unaffected — the latter closes under the *EH* assumed there, as Δ does — but the fixed-cut input is not available, so no claim is made outside $\theta' > 1/2$. That is the same threshold the no-go of [7] turns on, and not by accident: below it the cofactor left by the completion is no longer short.

The exchange, the split of Ξ and the shortness of its variable were verified as identities rather than as estimates, over $N = 2 \cdot 10^5$ to $3.2 \cdot 10^6$ at $\alpha = N^{0.44}$: the two sides of each agree to $2.274 \cdot 10^{-18}$ and $4.547 \cdot 10^{-18}$ relative to N, accumulated in index orders that share nothing, and the largest m carrying a surviving term is 213, 291, 393, 533, 727 against $\alpha =$ 215, 292, 396, 537, 728 (`audit_delta_bridge.py`).

8 Relation to the literature

This section is a placement, not a survey, and it is deliberately conservative about what is offered here as new.

Theorem 1. The mechanism is standard. After the divisor switch the Möbius on the long variable squares away and the surviving Möbius sits on a variable of length at most $N^{1/2-\delta}$, which is the classical configuration in which Bombieri–Vinogradov applies. Every ingredient (Bombieri–Vinogradov [6]; the Goldston–Yıldırım estimate [1] as used in [2]; the elementary density identity of Lemma 11) is off the shelf. The statement is offered as a *lemma about the Huang–Li reduction*, not as a contribution to the theory of Bombieri–Vinogradov; its content is that this particular consumption of EH_μ is unnecessary, and what difficulty it has lies in the bookkeeping — in particular that the condition $(k, N) = 1$ must be discarded by the degeneracy argument (Lemma 8) rather than expanded over $e \mid N$, and that main terms may be assigned only to classes with $(q, N) = 1$, the second of which shifts the density by exactly $N/\varphi(N)$ if missed (Remark 9).

Theorem 3. We are not aware of the identity in the literature. Its ingredients are again classical; what it does is locate, exactly, where the Goldbach content of the Huang–Li reduction sits — in the weight $\log k$, by $\mu * \log = \Lambda$.

Beyond the square-root barrier. Since [2] appeared, Lichtman [4] has shown that the primes have level of distribution $66/107 \approx 0.617$ using triply well-factorable weights, and has used it to improve upper bounds for Goldbach representations — the first use of a level beyond the square-root barrier on that problem. That work does not collide with the present one: it concerns the primes rather than $\sum_{n<N} \Lambda(n)\mu(N-n)$, and it yields upper bounds rather than the asymptotic that [2] targets. It is nevertheless the state of the art against which any claim of the form "beyond 1/2" must now be measured.

A level of distribution 3/5 for the Möbius function with triply well-factorable weights belongs to the earlier part of that line of work, [3]; we cite it here as the object the question below refers to rather than as a result we have checked against the published version. It is *not* in [4], which does not treat the Möbius function at all — the 3/5 appearing there is Maynard's prior record for the *primes*, which [4] improves to 66/107. Since [2] require $EH_\mu(N^{\theta'})$ for a single $\theta' > 1/2$, and since Theorem 3 identifies $w_k = \log k$ as the weight carrying the Goldbach content, the following question is well posed and, so far as we know, open:

> Does the weight that the Huang–Li consumption actually requires lie in the triply well-factorable class for which level $3/5$ is known for μ?

Two cautions attach to it. First, EH_μ as stated in (1) carries $\max_a |\cdot|$, whereas well-factorable results bound signed weighted sums $\sum_q \lambda_q E(x;q,a)$; the comparison must therefore be made against what the argument *consumes*, which is a signed sum, and not against the absolute-value form as written. Second, the no-go of [7] closes extraction *by divisor switching*; the technology of [4] is the Deshouillers–Iwaniec spectral large sieve, a different mechanism, and the no-go says nothing about it.

9 Summary

- Theorem 1: the Möbius-weighted, fixed-class correlation sum with weight $w_k = 1$ is $\ll_A N(\log N)^{-A}$, unconditionally, for any level $N^{\theta'}$ with $\theta' > 1/2$.
- Corollary 2: hence Huang–Li's E_4 (their Lemma 4) needs no hypothesis, and the whole EH_μ demand of their reduction collapses to the scalar E_3.
- Theorem 3: that scalar is unconditionally equivalent to Huang–Li's equation (22), hence already gives binary Goldbach for large even N, and gives the asymptotic $\tilde{r}(N) \sim \mathfrak{S}(N)N$ exactly when $C(N) = o(N)$.
- Propositions 5 and 6: Goldbach itself needs only a one-sided bound on E_3 at a threshold $\asymp N$, equivalently a constant-factor bound on the object EH_μ bounds — weaker than the consumed bound by a factor $(\log N)^A$ for every A.
- Proposition 20: the bound of Theorem 1 holds against the corrected formulation of [2], unconditionally for $\theta' > 1/2$. The two formulations differ by a term of Δ's shape, so the fixed cut is not superseded — it is one of the two ingredients and the omitted term is the other.
- Section 7: equation (18) of [2] omits an n-dependent constraint; the missing term Δ is exhibited and closes under hypotheses already assumed.
- Section 1.4: the net progress toward Goldbach is zero.

References

[1] D. Goldston and C. Yıldırım, *Higher correlations of divisor sums related to primes I*, Integers **3** (2003), A5.

[2] J.-J. Huang and H. Li, *On the connection between the Goldbach conjecture and the Elliott–Halberstam conjecture*, arXiv:2005.03811 [math.NT]; v1 (May 2020), v2 (August 2022). Also published in a Springer volume, `doi:10.1007/978-3-030-67996-5_17`. References here are to the arXiv version, which we have read, and not to the published version, which we have not.

[3] J. D. Lichtman, *Primes in arithmetic progressions to large moduli, II: well-factorable estimates*, arXiv:2006.07088 [math.NT], 2020.

[4] J. D. Lichtman (with an appendix by S. Drappeau), *Primes in arithmetic progressions to large moduli, and Goldbach beyond the square-root barrier*, arXiv:2309.08522 [math.NT], 2023.

[5] L. Mirsky, *The number of representations of an integer as the sum of a prime and a k-th power free integer*, Amer. Math. Monthly **56** (1949), 17–19.

[6] H. Montgomery and R. Vaughan, *Multiplicative number theory I: classical theory*, Cambridge University Press, 2007.

[7] *No weight extracts* $\sum_{n<N} \Lambda(n)\mu(N-n)$*: a no-go for the divisor-switch route, with the exact identities of its design space*, companion paper.

[8] C.-D. Pan, *A new attempt on Goldbach conjecture*, Chinese Ann. Math. **3** (1982), 555–560.

A Reproducibility

Every numerical figure printed above comes from one of the following scripts; each is standalone, runs under Python with numpy on a laptop, prints its own pass/fail criteria, and writes one result file.

Script	Result file	Supports
`audit_switch_identity.py`	`audit_switch_identity.txt`	(13)
`audit_delta_bridge.py`	`audit_delta_bridge.txt`	§7.1
`audit_threshold_arithmetic.py`	`audit_threshold_arithmetic.txt`	Note 19
`audit_density_identity.py`	`audit_density_identity.txt`	Lemma 11
`audit_E3_constant.py`	`audit_E3_constant.txt`	(22)
`lab_theta_sweep.py`	`lab_theta_sweep.txt`	Remark 15
`lab_onesided_margin.py`	`lab_onesided_margin.txt`	Measurement 17
`lab_onesided_demand.py`	`lab_onesided_demand.txt`	Measurement 18

No computation is used as a premise of any proof. The computations verify exact identities ((13), Lemma 11), fix the sign of a constant that the analysis determines ((22)), or measure a quantity whose size is stated only over the range measured.